From: Reka Halasz <reka.h@corvane-tx.example>
To: intake@[law firm]
Date: Mon, 2 Nov 2026 09:14
Subject: O-1 documents (everything I have)

Hi,

As discussed, here is everything I could find for the O-1. Sorry it is a bit of a mess. A few notes:

1. My current title is Head of Research at Corvane (I run the structure prediction work). HR's letter uses my old title, I think they have not updated it.

2. The Hungarian award certificate and the newspaper article from home are attached as they are. I do not have translations yet. My mum can translate them if that helps?

3. Prof. Ferrante sent me a draft of her letter first (forwarded below) and then the signed one. Both are attached, use whichever is better.

4. I was also named in BioScene Leaders' 40 Under 40 this year, which should help. The invoice for the honoree package is attached in case you need proof of it.

5. My passport is attached. I know it is getting close to the end, I will renew it next year.

6. Dr Brenner on our Scientific Advisory Board kindly wrote the "advisory opinion" letter you mentioned.

FoldSight now has over 5,000 GitHub stars and my papers have over 1,500 citations, so hopefully that is enough!

Thanks so much,
Reka

---------- Forwarded message ----------
From: Prof. Lucia Ferrante
Date: Fri, 2 Oct 2026 17:40
Subject: Draft letter for your petition, please check facts

Dear Reka, here is a first draft. Please check the facts before I sign. Lucia

> **DRAFT, NOT SIGNED**
>
> To USCIS: I am a professor of computational biology at Atlantic University. Dr Halasz's FoldSight method is used by 12 pharmaceutical companies and has become the standard tool for modelling binding pocket flexibility. It has been cited more than 2,000 times. I consider her one of the top five structural informaticians of her generation in the world.

---------- Forwarded message ----------
From: IJARS Editorial Office <editor@ijars-journal.example>
Date: 3 Mar 2025
Subject: Invitation to review for the International Journal of Advanced Research Science

> Dear Researcher, the International Journal of Advanced Research Science (IJARS) publishes original research in all fields of science, engineering, medicine, management and the arts. We welcome reviewers from every discipline. Accepted authors pay an article processing charge of USD 890. Reviewers receive a certificate of reviewing. Please reply within 24 hours.

(Reka: I did 2 reviews for them in 2025, certificate attached.)

---------- Forwarded message ----------
From: accounts@biosceneleaders.example
Date: 18 Jun 2026
Subject: Invoice BSL-2026-0412: 40 Under 40 Featured Honoree Package

Featured Honoree Package (profile, photo shoot, digital badge, 2 gala tickets)	USD 1,450.00
Status	PAID by card, 18 Jun 2026

Thank you for confirming your place on this year's 40 Under 40 in Biotech list.

Réka Halász, PhD

Cambridge, Massachusetts. Citizen of Hungary. Born 9 June 1991, Debrecen, Hungary.

Positions

Corvane Therapeutics, Inc., Cambridge, MA. Senior Research Scientist, then lead of the Structure Prediction group, February 2023 to present.

Alpine Institute of Technology, Zurich. Postdoctoral researcher, Laboratory of Prof. Elena Marchetti, October 2019 to January 2023.

Education

PhD in Computational Biology, Pannonia University of Sciences, Budapest, 2019. Thesis: "Learning the motion of protein binding pockets".
MSc in Biophysics, Pannonia University of Sciences, 2015.

Selected honours

- Protein Innovators Prize, International Society for Computational Structural Biology, 2025
- Fellow, European Society for Protein Informatics, elected 2025
- Fiatal Kutatói Díj (Young Researcher Award), Magyar Tudományos Társaság, 2020
- 40 Under 40 in Biotech, BioScene Leaders, 2026
- Spark Award, Corvane Therapeutics, 2024

Software

FoldSight (open source, 2022 to present), a tool that predicts how drug binding pockets in proteins change shape. More than 5,000 GitHub stars.

Publications (9)

1. **Halász R**, Marchetti E. FoldSight: learning binding pocket flexibility from molecular dynamics. *Journal of Structural Informatics* 18(3): 211 to 229, 2022.
2. Brandauer T, Ito K, Nwosu A, **Halász R**, Weller J, Marchetti E. Flexible pocket docking finds hidden binding sites across the human kinome. *Cell Systems Biology* 9(6): 744 to 761, 2024.
3. **Halász R**, Delgado M. Benchmarking pocket flexibility predictors on 1,200 drug targets. *Journal of Structural Informatics* 20(1): 33 to 49, 2024.
4. **Halász R**. Why drug discovery needs moving pictures, not snapshots (opinion). *The Scientist Weekly*, 12 May 2025.
5. Five further papers, 2016 to 2021, listed in full in the 2024 CV.

Citations: over 1,500 (Google Scholar).

Peer review

Reviewer, *Journal of Structural Informatics* (2024 to present).
Reviewer, *International Journal of Advanced Research Science* (2025).

Memberships

Fellow, European Society for Protein Informatics (2025).
Member, North American Protein Society (since 2019).

Reka Halasz-Toth, PhD

Senior Research Scientist, Corvane Therapeutics, Cambridge, MA (since February 2023).

Publications (7)

1. Halasz-Toth R, Marchetti E. FoldSight: learning binding pocket flexibility from molecular dynamics. J Struct Inform 2022.
2. Halasz-Toth R, Marchetti E, Varkonyi P. Pocket motion in GPCRs. Biophys Methods 2021.
3. Halasz-Toth R, Szabo L. Coarse grained pocket models. Hung J Biophys 2020.
4. Halasz-Toth R. Thesis summary. Pannonia Sci Rep 2019.
5. Halasz-Toth R, Kiss A. Conformational sampling benchmarks. Comput Biol Lett 2018.
6. Kiss A, Halasz-Toth R. Water networks in binding sites. Hung J Biophys 2017.
7. Halasz-Toth R. MSc project report. Pannonia Sci Rep 2016.

Citations: 640 (Google Scholar, March 2024). FoldSight: 2,100 GitHub stars.

MAGYARORSZÁG / HUNGARY ÚTLEVÉL / PASSPORT

Type / Típus	P	Code / Kód	HUN
Passport No. / Útlevélszám	BK4471902		
Surname / Családi név	HALÁSZ		
Given names / Utónév	RÉKA		
Nationality	HUNGARIAN / MAGYAR		
Date of birth	09 JUN 1991	Sex	F
Place of birth	DEBRECEN		
Date of issue	03 MAR 2017	Date of expiry	**02 MAR 2027**

P<HUNHALASZ<<REKA<<<<<<<<<<<<<<<<<<<<<<
BK44719021HUN9106092F2703022<<<<<<<<<<<<<<04

Most Recent I-94

U.S. Customs and Border Protection. Printed 28 Oct 2026.

Admission (I-94) Record Number	482911073A3
Most Recent Date of Entry	2024 January 19
Class of Admission	H1B
Admit Until Date	**01/15/2027**
Last/Surname	HALASZ
First (Given) Name	REKA
Birth Date	06/09/1991
Passport Number	BK4471902
Country of Issuance	Hungary

Employment Agreement

Between **Corvane Therapeutics, Inc.**, a Delaware corporation with offices at 400 Binney Street, Cambridge, MA 02142 (the "Company"), and **Réka Halász** (the "Employee"). Dated 30 October 2026.

1. Position

The Company will employ the Employee as Principal Research Scientist, Structure Prediction, reporting to the Chief Scientific Officer. The Employee will lead the Structure Prediction group of six scientists and will be responsible for the FoldSight platform and its use in the Company's drug discovery programmes.

2. Term

Employment under this Agreement begins on the later of 4 January 2027 or the date U.S. Citizenship and Immigration Services approves the Company's O-1 petition for the Employee, and continues for three years unless ended earlier under section 11.

3. Base salary

The Company will pay the Employee a base salary of USD 205,000 per year, paid in equal instalments on the Company's normal payroll dates, less lawful deductions. Salary will be reviewed each year in March.

4. Bonus

The Employee is eligible for an annual discretionary bonus with a target of 10 percent of base salary, based on Company and individual performance. Bonuses are not guaranteed.

5. Equity

Subject to approval by the Board, the Company will grant the Employee options to buy 40,000 shares of common stock under the 2021 Equity Incentive Plan, vesting over four years with a one year cliff.

6. Benefits

The Employee may take part in the Company's health, dental, vision and 401(k) plans on the same terms as other full time employees, and receives 25 days of paid time off each year.

7. Place of work

The Employee will work at the Company's offices at 400 Binney Street, Cambridge, Massachusetts, and may travel to conferences, collaborators and partner sites as described in the attached itinerary.

8. Duties and conduct

The Employee will devote full working time to the Company, follow the Company's policies, and not take on other paid work without written consent, except unpaid academic peer review and conference service.

9. Confidentiality and inventions

The Employee will keep the Company's confidential information secret during and after employment. Inventions made in the course of employment belong to the Company. Open source contributions to FoldSight continue under the project's existing licence, as approved by the Company on 1 March 2023.

10. Immigration

The Company will file and pay for an O-1 petition and any extension needed for the Employee to perform this role. The Employee will provide accurate documents promptly when asked.

11. Ending employment

Either party may end employment with 60 days' written notice. If the Company ends employment without cause, it will pay three months' base salary and the reasonable cost of return transport to Hungary, as required for O-1 workers.

12. Whole agreement

This Agreement replaces the Employee's offer letter of 9 January 2023 and the role description in it. It is governed by the laws of Massachusetts.

Signed for the Company: *Dr. Andrew Kessler*, Chief Scientific Officer, 30 October 2026
Signed by the Employee: *Réka Halász*, 30 October 2026

Itinerary: Réka Halász, O-1 period January 2027 to January 2030

Dates	Activity	Location
Throughout	Lead the Structure Prediction group; FoldSight platform development	Corvane, 400 Binney Street, Cambridge, MA
Quarterly from Q1 2027	Joint programme reviews with Northgate Pharma	Northgate Pharma, Princeton, NJ
14 to 18 June 2027	Invited talk, ISCSB annual meeting	Montreal, Canada (travel abroad)
September 2027	FoldSight 3.0 public release	Cambridge, MA
Annually	European Society for Protein Informatics fellows' meeting	Varies (Europe)

Signed: Dr. Andrew Kessler, Chief Scientific Officer, 30 October 2026

CORVANE THERAPEUTICS, INC. | People Team | 400 Binney Street, Cambridge, MA 02142

26 October 2026

To whom it may concern,

This letter confirms that Dr. Reka Halasz has been employed full time by Corvane Therapeutics, Inc. since 6 February 2023. Her current job title is **Senior Research Scientist**. She leads the Structure Prediction group, which has five scientists. Her current base salary is USD 185,000 per year.

Corvane has employed her in H-1B status throughout her employment.

Yours faithfully,
Maria Chen, Head of People

Dr. Paul Brenner
Member, Scientific Advisory Board, Corvane Therapeutics, Inc.
12 October 2026

To USCIS,

I write as a member of Corvane's Scientific Advisory Board, where I have advised the company since 2022 and hold a small number of Corvane shares. I have known Dr Halász since she joined Corvane. I fully support the petition. She is an outstanding scientist and her work is essential to the company. I have no objection to the grant of O-1 status.

Dr. Paul Brenner

Corvane Therapeutics: company profile (from the company's investor pack, September 2026)

Founded 2020. Cambridge, Massachusetts. 85 employees as of 1 September 2026. Uses machine learning on protein motion to find new drug binding sites. Pipeline: two oncology programmes in preclinical development.

Funding: Seed USD 9M (2020); Series A USD 24M (2022); Series B USD 60M (May 2025, led by Harborline Bio Ventures).

Partnerships: multi year research collaboration with Northgate Pharma (signed 2025).

Biotech Ledger, 3 February 2026: "Ten AI drug discovery startups to watch"

... 7. Corvane Therapeutics (Cambridge, MA). Corvane's bet is that drugs fail because we model proteins as statues. Its open source FoldSight tool, built by its structure prediction lead Réka Halász, is used widely in academia, and the company has signed a research deal with Northgate.

About Biotech Ledger: a daily online publication covering the biotechnology industry, founded 2016, 210,000 newsletter subscribers (publisher's figure).

Corvane names Réka Halász Principal Research Scientist

Cambridge, MA, 29 October 2026. Corvane Therapeutics, a company of over 100 people, today announced that Dr. Réka Halász will become Principal Research Scientist from January 2027. "Réka is one of the brightest minds in computational biology," said CEO Laura Mendez. "Her work is the foundation of everything we do."

Media contact: press@corvane.example

CORVANE SPARK AWARD 2024

Presented to Réka Halász

For the most helpful code review culture in the company.

Chosen by colleague vote. Signed, Laura Mendez, CEO

4 August 2025

Dear Dr Halász,

The ISCSB Awards Committee is pleased to tell you that you are one of the three recipients of the 2025 Protein Innovators Prize, for the development of FoldSight. The prize will be presented at the ISCSB annual meeting in Lisbon on 22 September 2025 and carries an award of EUR 20,000.

Prof. Hannah Lindqvist, Chair, ISCSB Awards Committee

ISCSB announces 2025 Protein Innovators Prize winners

The International Society for Computational Structural Biology has about 4,200 members in 58 countries. The Protein Innovators Prize, awarded each year since 2009, recognises researchers within 15 years of their doctorate whose methods have changed practice in the field.

This year the committee received 612 nominations from 41 countries. Nominations were reviewed by a panel of 12 senior scientists from universities and industry in Europe, North America and Asia. Panel members do not review nominees from their own institution.

The 2025 winners are Dr Réka Halász (Corvane Therapeutics, USA), Dr Tomás Arriaga (University of Santiago, Chile) and Dr Mei-Ling Hsu (National Institute of Molecular Sciences, Taiwan).

Photograph caption: Dr Halász receives the prize in Lisbon, 22 September 2025.

MAGYAR TUDOMÁNYOS TÁRSASÁG

OKLEVÉL

A Magyar Tudományos Társaság Elnöksége

Halász Réka

részére

FIATAL KUTATÓI DÍJAT

adományoz a fehérjék kötőhelyeinek mozgását leíró számítási módszerek kidolgozásáért.

A díjat évente legfeljebb öt, 35 év alatti magyar kutató kaphatja, országos pályázat alapján, a Társaság *tudományos bizottságának javaslatára.*

Budapest, 2020. november 12.

Prof. Dr. Szentgyörgyi Ilona, elnök

HAJDÚSÁGI NAPLÓ

2020. november 14., szombat | Helyi hírek, 5. oldal

Debreceni kutatónő kapta a Fiatal Kutatói Díjat

Írta: Nagy Péter

Halász Réka, a debreceni Fazekas Gimnázium egykori diákja, elnyerte a Magyar Tudományos Társaság Fiatal Kutatói Díját. A 29 éves kutató jelenleg Zürichben dolgozik, ahol azt vizsgálja, hogyan változtatja alakját a fehérjék felszíne, amikor egy gyógyszermolekula hozzájuk kapcsolódik.

"Nagyon büszkék vagyunk rá" mondta egykori osztályfőnöke, Tóth Erzsébet. A díjat Budapesten adták át csütörtökön.

Réka Halász: the protein whisperer

Featured honoree. Réka Halász is redefining drug discovery at Corvane Therapeutics with FoldSight, used by thousands of scientists worldwide. "Never stop asking how things move," she says.

BioScene Leaders celebrates rising stars across the life sciences. Honorees are selected from nominations, including self nominations. Featured Honoree packages are available.

20 May 2025

Dear Dr Halász,

I am delighted to tell you that the Council of the European Society for Protein Informatics has elected you a Fellow of the Society (FESPI), on the recommendation of the Fellowship Committee. Your election will be announced at the General Assembly in Heidelberg on 30 June 2025.

Prof. Jan de Wit, President

ESPI Bylaws (extract)

Article 4. Classes of membership

4.1 Member. Any person working or studying in protein science or a related field may join as a Member by paying the annual subscription.

4.2 Associate Member. Students may join as Associate Members at a reduced subscription.

4.3 Fellow. Fellowship is the Society's highest class of membership and is conferred by election only.

Article 7. Election of Fellows

7.1 A candidate for Fellowship must be nominated by two existing Fellows who are not employed by the same organisation as the candidate.

7.2 Nominations are assessed by the Fellowship Committee, made up of nine senior scientists elected by the Fellows for a term of three years.

7.3 The Committee may recommend a candidate only if it finds that the candidate has made **outstanding and sustained contributions to protein informatics**, as shown by published work, widely used methods or tools, and recognition by peers.

7.4 No more than fifteen Fellows may be elected in any calendar year.

7.5 Fellowship cannot be applied for or purchased.

Article 9. Subscriptions

9.1 Members pay an annual subscription set by Council. Fellows pay the same subscription as Members.

Article 10. Council

10.1 The Society is governed by a Council of twelve members elected by the Members for three years.

10.2 The Council may make standing orders under this Article.

Article 11. General Assembly

11.1 A General Assembly of Members is held at least once in each calendar year.

11.2 The Council may make standing orders under this Article.

Article 12. Finance

12.1 The financial year runs from 1 January to 31 December. Accounts are audited each year.

12.2 The Council may make standing orders under this Article.

Article 13. Prizes

13.1 The Society may award prizes for outstanding work by early career researchers.

13.2 The Council may make standing orders under this Article.

Article 14. Ethics

14.1 Members and Fellows must follow the Society's code of research conduct.

14.2 The Council may make standing orders under this Article.

Article 15. Amendments

15.1 These Bylaws may be amended by a two thirds majority of the General Assembly.

15.2 The Council may make standing orders under this Article.

NORTH AMERICAN PROTEIN SOCIETY

Member since 2019: Reka Halasz

Membership is open to anyone interested in protein science. Join online at any time. Annual dues USD 95.

PANNÓNIA TUDOMÁNYEGYETEM

DOKTORI OKLEVÉL

Halász Réka részére, aki a doktori fokozatot (PhD) számítógépes biológia tudományágban summa cum laude minősítéssel megszerezte.

Budapest, 2019. június 28.

CERTIFIED TRANSLATION

PANNONIA UNIVERSITY OF SCIENCES. DOCTORAL DIPLOMA. Awarded to Réka Halász, who has obtained the doctoral degree (PhD) in the discipline of computational biology with the grade summa cum laude. Budapest, 28 June 2019.

I, Gábor Fekete, certify that I am competent to translate from Hungarian into English and that the above is a complete and accurate translation of the attached document. Gábor Fekete, 15 October 2026. Address: 12 Elm Street, Somerville, MA.

BOSTON LIFE SCIENCES REVIEW

Tuesday 14 April 2026 | Section B: Research | Page B4

The scientist who taught algorithms to see proteins breathe

By Dana Whitfield, Senior Science Correspondent

Most computer models treat a protein like a statue. Réka Halász thinks that is why so many drug candidates fail. The Hungarian born scientist, who leads structure prediction at Cambridge start up Corvane Therapeutics, built FoldSight, a free tool that predicts how the pockets where drugs attach open and close.

Since its release in 2022, FoldSight has been downloaded by research groups on every continent, according to its public repository. Last year an international society of structural biologists gave Halász its Protein Innovators Prize.

"Proteins move. If you ignore that, you are designing keys for a lock that is not there," Halász said in her office overlooking the Charles River.

Dr. Mark Delaney of Northgate Pharma said his team used FoldSight to find a binding site in a cancer target that earlier methods had missed. "It changed the direction of the programme," he said.

Boston Life Sciences Review: Media Kit 2026

Published weekly since 1994 in print and online. Covers biotechnology, pharmaceuticals and life science research in New England and nationally.

Print circulation (audited, 2025)	38,500
Online monthly unique visitors	410,000
Editorial staff	31 full time journalists
Readership	Scientists, executives and investors in life sciences

Why drug discovery needs moving pictures, not snapshots

By Réka Halász. Dr Halász is a scientist at Corvane Therapeutics.

For twenty years our industry has designed drugs against frozen pictures of proteins. Crystal structures are beautiful, but a protein in the body is never still. Binding pockets breathe, open and close. When we ignore this motion we miss targets, and we misjudge the ones we find.

The good news is that machine learning trained on molecular dynamics now lets us predict this motion cheaply. My colleagues and I have released our methods openly, because the field moves faster when everyone can test the same tools.

Regulators, funders and companies should ask a simple question of every structure based programme: did we look at how the target moves?

JOURNAL OF STRUCTURAL INFORMATICS | Editorial Office

3 October 2026

To whom it may concern,

This confirms that Dr Réka Halász has served as an invited peer reviewer for the Journal of Structural Informatics since March 2024 and has completed reviews of 11 manuscripts. Reviewers are invited by the handling editor on the basis of their published expertise. The journal received 1,960 submissions in 2025 and accepted 21 percent of them.

Prof. Ravi Menon, Editor in Chief

INTERNATIONAL JOURNAL OF ADVANCED RESEARCH SCIENCE

CERTIFICATE OF REVIEWING

Awarded to Reka Halasz in recognition of outstanding contribution to the quality of the journal, 2025.

Prof. Lucia Ferrante
Department of Computational Biology, Atlantic University, Lisbon
8 October 2026

To USCIS,

I am a professor of computational biology with over 20 years of research on protein dynamics and about 160 publications. I have never worked with Dr Halász, Corvane or her former laboratory. I know her work from the literature and from using FoldSight in my own group.

Before FoldSight, predicting how a binding pocket changes shape required weeks of simulation for each protein. FoldSight made that prediction possible in minutes, and it was tested openly on more than a thousand drug targets. My group and at least four others I know of now use it routinely, and I am aware that several pharmaceutical companies use it. I regard it as one of the most useful method contributions to our field in recent years, and Dr Halász as one of the leading scientists of her generation in protein informatics.

Signed: Lucia Ferrante

Dr. Mark Delaney
Vice President, Computational Chemistry, Northgate Pharma, Princeton, NJ
15 October 2026

To USCIS,

Northgate is a pharmaceutical company with about 9,000 employees. My team began using the open source FoldSight tool in 2023, before any relationship with Corvane. Using it we identified a previously unknown binding site in an oncology target; a candidate drug designed against that site entered a Phase 1 clinical trial in 2026. FoldSight has since been used in three of our discovery programmes.

In 2025 Northgate signed a paid research collaboration with Corvane, partly because of this work. I am writing because Dr Halász's method made a real difference to our pipeline.

Signed: Mark Delaney

Prof. Elena Marchetti
Alpine Institute of Technology, Zurich
1 October 2026

To USCIS,

Dr Halász was a postdoctoral researcher in my laboratory from 2019 to 2023, and I am co-author of the FoldSight paper. She conceived the core idea of FoldSight and wrote most of its code; my role was supervision and funding. She is the most creative researcher I have supervised in 25 years.

Signed: Elena Marchetti

Dr. Andrew Kessler
Chief Scientific Officer, Corvane Therapeutics, Inc.
30 October 2026

To USCIS,

Dr Halász leads our Structure Prediction group. Both of our preclinical programmes started from binding sites her group found with FoldSight, and our collaboration with Northgate Pharma depends on her platform. Losing her would stop that work. From January 2027 she will be Principal Research Scientist.

Signed: Andrew Kessler

github.com/foldsight/foldsight

Public repository. Printed 1 November 2026.

Stars	4,812
Forks	911
Contributors	63
First release	v1.0, 14 March 2022
Latest release	v2.6, 2 October 2026
Top contributor by commits	rhalasz (2,140 commits, 71 percent)
Package downloads (last 12 months, package index)	182,000

Réka Halász

Corvane Therapeutics. Verified email at corvane.example

protein dynamics drug discovery machine learning

	All	Since 2021
Citations	**1,418**	1,301
h-index	14	13
i10-index	16	15

Title	Cited by	Year
FoldSight: learning binding pocket flexibility from molecular dynamics	612	2022
Flexible pocket docking finds hidden binding sites across the human kinome	389	2024
Benchmarking pocket flexibility predictors on 1,200 drug targets	141	2024
Pocket motion in GPCRs	97	2021

Printed 1 Nov 2026

Journal of Structural Informatics 18(3): 211 to 229 (2022). Peer reviewed. Published by the Society for Structural Informatics.

FoldSight: learning binding pocket flexibility from molecular dynamics

Réka Halász[1], Elena Marchetti[1]
1 Alpine Institute of Technology, Zurich

Abstract

We present FoldSight, a graph neural network trained on 14,000 molecular dynamics trajectories that predicts the range of shapes a protein binding pocket can take, from a single static structure, in under five minutes on one graphics card. On a held out set of 310 proteins FoldSight recovers 83 percent of cryptic pockets seen in long simulations, compared with 41 percent for the best previous method. We release FoldSight as open source software.

1. Introduction

Binding pockets are not fixed. Molecular dynamics simulations show that many pockets open, close and change shape on time scales from nanoseconds to milliseconds, and that some binding sites exist only in conformations that are rarely seen in crystal structures. Structure based drug design has mostly ignored this motion because simulating it for every target is expensive: a single microsecond scale simulation of a medium sized protein can take weeks of computing time.

Binding pockets are not fixed. Molecular dynamics simulations show that many pockets open, close and change shape on time scales from nanoseconds to milliseconds, and that some binding sites exist only in conformations that are rarely seen in crystal structures. Structure based drug design has mostly ignored this motion because simulating it for every target is expensive: a single microsecond scale simulation of a medium sized protein can take weeks of computing time.

2. Methods

Training data were 14,000 trajectories from public databases and our own simulations, covering 2,900 proteins. Each frame was converted to a residue graph. The network predicts, for each residue lining a pocket, a distribution over side chain and backbone displacements. Pocket shapes are sampled from these distributions and clustered.

Binding pockets are not fixed. Molecular dynamics simulations show that many pockets open, close and change shape on time scales from nanoseconds to milliseconds, and that some binding sites exist only in conformations that are rarely seen in crystal structures. Structure based drug design has mostly ignored this motion because simulating it for every target is expensive: a single microsecond scale simulation of a medium sized protein can take weeks of computing time.

We trained the network in three stages. First, residue embeddings were pre trained on static structures to predict local geometry. Second, the model was trained on trajectory frames to predict displacement distributions for pocket lining residues. Third, it was fine tuned with a contrastive loss that rewards sampling pocket shapes seen in long simulations but absent from crystal structures. Hyperparameters were selected on a validation set of 240 proteins that shared less than 30 percent sequence identity with any training protein. All results reported here are on a separate test set of 310 proteins.

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3. Results

Method	Cryptic pockets recovered	Median run time
FoldSight	83%	4.2 min
PocketFlex	41%	38 min

MDpocket short runs	36%	31 h
Static docking	12%	2 min

FoldSight recovered 83 percent of cryptic pockets (95 percent confidence interval 79 to 87) against 41 percent for the previous best method. Median run time was 4.2 minutes per protein.

Binding pockets are not fixed. Molecular dynamics simulations show that many pockets open, close and change shape on time scales from nanoseconds to milliseconds, and that some binding sites exist only in conformations that are rarely seen in crystal structures. Structure based drug design has mostly ignored this motion because simulating it for every target is expensive: a single microsecond scale simulation of a medium sized protein can take weeks of computing time.

Author contributions

R.H. conceived the method, wrote the software, ran all experiments and wrote the paper. E.M. supervised the work and edited the paper.

4. Discussion

We trained the network in three stages. First, residue embeddings were pre trained on static structures to predict local geometry. Second, the model was trained on trajectory frames to predict displacement distributions for pocket lining residues. Third, it was fine tuned with a contrastive loss that rewards sampling pocket shapes seen in long simulations but absent from crystal structures. Hyperparameters were selected on a validation set of 240 proteins that shared less than 30 percent sequence identity with any training protein. All results reported here are on a separate test set of 310 proteins.

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References

1. Olsen B, Silva C, Rossi D. Advances in protein language models. *Mol Dyn Rev* 35: 885, 2014.
2. Fischer C, Zhang D, Quinn E. Advances in protein language models. *Biophys J Comput* 9: 467, 2017.
3. Eriksen D, Costa E, Rossi F. Advances in cryptic pocket detection. *Bioinform Adv* 31: 679, 2013.
4. Tanaka E, Abe F, Novak G. Advances in free energy perturbation. *Proteins Comput Methods* 6: 46, 2014.
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Flexible pocket docking finds hidden binding sites across the human kinome

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Abstract

Using flexible pocket models for all 518 human kinases, we find 92 previously unreported binding sites, of which 17 were confirmed by crystallography. Several are now the subject of drug discovery programmes.

Introduction

Binding pockets are not fixed. Molecular dynamics simulations show that many pockets open, close and change shape on time scales from nanoseconds to milliseconds, and that some binding sites exist only in conformations that are rarely seen in crystal structures. Structure based drug design has mostly ignored this motion because simulating it for every target is expensive: a single microsecond scale simulation of a medium sized protein can take weeks of computing time.

Results

Binding pockets are not fixed. Molecular dynamics simulations show that many pockets open, close and change shape on time scales from nanoseconds to milliseconds, and that some binding sites exist only in conformations that are rarely seen in crystal structures. Structure based drug design has mostly ignored this motion because simulating it for every target is expensive: a single microsecond scale simulation of a medium sized protein can take weeks of computing time.

Methods

We trained the network in three stages. First, residue embeddings were pre trained on static structures to predict local geometry. Second, the model was trained on trajectory frames to predict displacement distributions for pocket lining residues. Third, it was fine tuned with a contrastive loss that rewards sampling pocket shapes seen in long simulations but absent from crystal structures. Hyperparameters were selected on a validation set of 240 proteins that shared less than 30 percent sequence identity with any training protein. All results reported here are on a separate test set of 310 proteins.

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Author contributions

T.B. and E.M. designed the study. T.B., K.I. and A.N. ran the docking screens. R.H. adapted FoldSight for kinases and analysed the pocket motion results. J.W. performed crystallography. T.B. wrote the paper with input from all authors.

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Benchmarking pocket flexibility predictors on 1,200 drug targets

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Abstract

We compare seven pocket flexibility predictors on 1,200 approved and clinical drug targets. FoldSight and two newer methods built on it perform best. We release the benchmark so that future methods can be compared fairly.

Author contributions

R.H. designed the benchmark and wrote the paper. M.D. ran three of the seven methods.

1. Introduction

We trained the network in three stages. First, residue embeddings were pre trained on static structures to predict local geometry. Second, the model was trained on trajectory frames to predict displacement distributions for pocket lining residues. Third, it was fine tuned with a contrastive loss that rewards sampling pocket shapes seen in long simulations but absent from crystal structures. Hyperparameters were selected on a validation set of 240 proteins that shared less than 30 percent sequence identity with any training protein. All results reported here are on a separate test set of 310 proteins.

2. Methods

We trained the network in three stages. First, residue embeddings were pre trained on static structures to predict local geometry. Second, the model was trained on trajectory frames to predict displacement distributions for pocket lining residues. Third, it was fine tuned with a contrastive loss that rewards sampling pocket shapes seen in long simulations but absent from crystal structures. Hyperparameters were selected on a validation set of 240 proteins that shared less than 30 percent sequence identity with any training protein. All results reported here are on a separate test set of 310 proteins.

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Method	Cryptic pockets recovered	Median run time
FoldSight	83%	4.2 min
PocketFlex	41%	38 min
MDpocket short runs	36%	31 h
Static docking	12%	2 min

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Kowalski et al.	Applying FoldSight to protein language models	*Bioinform Adv*	2022
Fischer et al.	Applying FoldSight to allosteric site prediction	*Cell Syst Biol*	2022
Wagner et al.	Applying FoldSight to allosteric site prediction	*Cell Syst Biol*	2023
Xu et al.	Applying FoldSight to allosteric site prediction	*Drug Des Lett*	2024
Zhang et al.	Applying FoldSight to kinase inhibitor selectivity	*Nucleic Acids Comput*	2026
Lee et al.	Applying FoldSight to molecular dynamics sampling	*Drug Des Lett*	2023
Jensen et al.	Applying FoldSight to graph neural networks for proteins	*J Chem Inf Model Res*	2024
Eriksen et al.	Applying FoldSight to molecular dynamics sampling	*Mol Dyn Rev*	2022
Olsen et al.	Applying FoldSight to free energy perturbation	*Proteins Comput Methods*	2026
Park et al.	Applying FoldSight to cryptic pocket detection	*J Struct Inform*	2026
Schmidt et al.	Applying FoldSight to kinase inhibitor selectivity	*Drug Des Lett*	2024
Nakamura et al.	Applying FoldSight to binding site druggability	*Proteins Comput Methods*	2023
Jensen et al.	Applying FoldSight to allosteric site prediction	*J Chem Inf Model Res*	2023
Ivanova et al.	Applying FoldSight to free energy perturbation	*J Struct Inform*	2023
Zhang et al.	Applying FoldSight to graph neural networks for proteins	*J Struct Inform*	2026
Olsen et al.	Applying FoldSight to binding site druggability	*Bioinform Adv*	2022
Park et al.	Applying FoldSight to protein language models	*J Struct Inform*	2023
Costa et al.	Applying FoldSight to free energy perturbation	*Nucleic Acids Comput*	2024
Varga et al.	Applying FoldSight to protein language models	*J Struct Inform*	2026
Jensen et al.	Applying FoldSight to free energy perturbation	*Drug Des Lett*	2026
Fischer et al.	Applying FoldSight to binding site druggability	*Biophys J Comput*	2022
Tanaka et al.	Applying FoldSight to GPCR conformations	*J Chem Inf Model Res*	2025
Rossi et al.	Applying FoldSight to kinase inhibitor selectivity	*J Chem Inf Model Res*	2024

Petrov et al.	Applying FoldSight to free energy perturbation	*J Chem Inf Model Res*	2026
Tanaka et al.	Applying FoldSight to protein language models	*Cell Syst Biol*	2022
Garcia et al.	Applying FoldSight to binding site druggability	*J Struct Inform*	2026
Olsen et al.	Applying FoldSight to graph neural networks for proteins	*Nucleic Acids Comput*	2023
Tanaka et al.	Applying FoldSight to cryptic pocket detection	*Mol Dyn Rev*	2025
Quinn et al.	Applying FoldSight to protein language models	*Cell Syst Biol*	2022
Costa et al.	Applying FoldSight to molecular dynamics sampling	*Biophys J Comput*	2026
Ueda et al.	Applying FoldSight to free energy perturbation	*J Chem Inf Model Res*	2026
Eriksen et al.	Applying FoldSight to cryptic pocket detection	*Cell Syst Biol*	2025
Bauer et al.	Applying FoldSight to ensemble docking	*Drug Des Lett*	2024
Young et al.	Applying FoldSight to binding site druggability	*Mol Dyn Rev*	2022
Ueda et al.	Applying FoldSight to free energy perturbation	*Biophys J Comput*	2025
Ivanova et al.	Applying FoldSight to allosteric site prediction	*Biophys J Comput*	2025
Wagner et al.	Applying FoldSight to binding site druggability	*Mol Dyn Rev*	2022
Kowalski et al.	Applying FoldSight to GPCR conformations	*Mol Dyn Rev*	2024
Wagner et al.	Applying FoldSight to allosteric site prediction	*J Struct Inform*	2023
Petrov et al.	Applying FoldSight to binding site druggability	*Biophys J Comput*	2023
Xu et al.	Applying FoldSight to protein language models	*Drug Des Lett*	2022
Abe et al.	Applying FoldSight to allosteric site prediction	*Bioinform Adv*	2026
Eriksen et al.	Applying FoldSight to molecular dynamics sampling	*Bioinform Adv*	2026
Jensen et al.	Applying FoldSight to protein language models	*Bioinform Adv*	2022
Bauer et al.	Applying FoldSight to binding site druggability	*J Chem Inf Model Res*	2024
Ivanova et al.	Applying FoldSight to molecular dynamics sampling	*Drug Des Lett*	2023
Nakamura et al.	Applying FoldSight to protein language models	*Proteins Comput Methods*	2023
Lambert et al.	Applying FoldSight to kinase inhibitor selectivity	*Nucleic Acids Comput*	2026
Garcia et al.	Applying FoldSight to cryptic pocket detection	*Biophys J Comput*	2025
Schmidt et al.	Applying FoldSight to cryptic pocket detection	*Nucleic Acids Comput*	2026
Wagner et al.	Applying FoldSight to binding site druggability	*Biophys J Comput*	2025
Silva et al.	Applying FoldSight to GPCR conformations	*J Chem Inf Model Res*	2025
Moreau et al.	Applying FoldSight to cryptic pocket detection	*J Chem Inf Model Res*	2025
Lambert et al.	Applying FoldSight to allosteric site prediction	*Biophys J Comput*	2026
Zhang et al.	Applying FoldSight to graph neural networks for proteins	*Mol Dyn Rev*	2023

Moreau et al.	Applying FoldSight to allosteric site prediction	*Mol Dyn Rev*	2024
Petrov et al.	Applying FoldSight to protein language models	*Mol Dyn Rev*	2023
Silva et al.	Applying FoldSight to protein language models	*J Chem Inf Model Res*	2026
Moreau et al.	Applying FoldSight to GPCR conformations	*Proteins Comput Methods*	2022
Wagner et al.	Applying FoldSight to kinase inhibitor selectivity	*Nucleic Acids Comput*	2024
Tanaka et al.	Applying FoldSight to cryptic pocket detection	*Drug Des Lett*	2023
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Young et al.	Applying FoldSight to free energy perturbation	*J Struct Inform*	2024
Moreau et al.	Applying FoldSight to binding site druggability	*J Chem Inf Model Res*	2025

Corvane Therapeutics

Drugs fail because we model proteins as statues. We model them as they move.

Confidential. Corvane Therapeutics. September 2026.

The problem

Around 90 percent of drug candidates that enter clinical trials fail. Many binding sites only exist when the protein moves.

Our platform

FoldSight, developed by our structure prediction lead, predicts pocket motion in minutes. Open source core, proprietary pipeline on top.

Traction

Two preclinical oncology programmes. Research collaboration with Northgate Pharma (2025). Over 100 team members.

Team

CEO Laura Mendez (ex Northgate). CSO Andrew Kessler. Head of Research: Réka Halász.

Financing

Raised USD 93M to date. Series B USD 60M led by Harborline Bio Ventures, May 2025.

Form W-2 Wage and Tax Statement 2025 Copy B, to be filed with employee's federal tax return

a Employee's social security number	XXX-XX-4417
b Employer identification number	84-2219034
c Employer's name and address	Corvane Therapeutics, Inc., 400 Binney Street, Cambridge, MA 02142
e Employee's name	Reka Halasz
1 Wages, tips, other compensation	191,212.40
2 Federal income tax withheld	36,840.11
12a Code D (401(k) deferral)	23,500.00

Box 1 includes base salary and the 2024 annual bonus paid in March 2025.

Corvane Therapeutics: Earnings Statement

Employee: Reka Halasz Pay period: 1 to 31 October 2026 Pay date: 31 October 2026

Earnings	Rate	Current	Year to date
Regular salary	185,000.00 / year	15,416.67	154,166.70
Bonus		0.00	17,250.00

Dr. Mark Delaney
Vice President, Computational Chemistry, Northgate Pharma, Princeton, NJ
15 October 2026

To USCIS,

Northgate is a pharmaceutical company with about 9,000 employees. My team began using the open source FoldSight tool in 2023, before any relationship with Corvane. Using it we identified a previously unknown binding site in an oncology target; a candidate drug designed against that site entered a Phase 1 clinical trial in 2026. FoldSight has since been used in three of our discovery programmes.

In 2025 Northgate signed a paid research collaboration with Corvane, partly because of this work. I am writing because Dr Halász's method made a real difference to our pipeline.

Signed: Mark Delaney